

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
MID SEMESTER EXAMINATION, 2023
SESSION: 2023 – 2024 (Autumn)

Subject code: CS3009
No. of pages: 2

Subject Name: Operating Systems
Full Marks: 30

Dept. Code: CS
Duration: 2 Hrs

Note: Answer all Questions. Questions can be answered in any serial order. However, ensure that all parts of a question are answered sequentially.

1. (a) What are the primary and secondary goals of an OS? Explain with reasoning. [2]
(b) Differentiate between Zombie and Orphan Process. [2]
(c) Differentiate between Distributed and Network Operating Systems with diagrams. Which one among these is called a tightly coupled system. [2]
(d) List the different Process States in OS. How many states a process must pass through in its life-cycle to complete? Where do these states reside in memory? [2]
(e) Differentiate between Long-term scheduler and Short-term scheduler. What is the job of a Dispatcher? [2]
2. (a) What is Cascaded Termination? In which condition does it occur? [2]
(b) Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. Compute the percentage of time the CPU remain idle. [4]
(c) The following program consists of 3 concurrent processes and 3 binary semaphores. The semaphores are initialized as $S_0 = 1$, $S_1 = 0$, $S_2 = 0$. [4]

Process P_0	Process P_1	Process P_2
<pre>while (true){ wait (S_0); print (0); signal (S_1); signal (S_2); }</pre>	<pre>wait (S_1); signal (S_0);</pre>	<pre>wait (S_2); signal (S_0);</pre>

How many *minimum* and *maximum* times will process P_0 print '0'? Assume that the process P_1 and P_2 run exactly once.

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3. (a) What is Race Condition? Two processes, P1 and P2, need to access a critical section of code. Consider the following synchronization construct used by the processes: [5]

Process P1	Process P2
<pre>while (true) { wants1 = true; while (wants2 == true); /* Critical Section */ wants1=false; }</pre>	<pre>while (true) { wants2 = true; while (wants1==true); /* Critical Section */ wants2 = false; }</pre>

The initial values of shared boolean variables wants1 and wants2 are randomly assigned. Examine and explain the feasibility of the solution w.r.t all the required criteria. Also, explain if the deadlock is possible while processes try to access the critical section of the code.

- (b) Consider the set of five processes whose arrival time and burst time are given below: [3]

Process	Arrival Time	Burst Time
P ₁	0	5
P ₂	1	3
P ₃	2	1
P ₄	3	2
P ₅	4	3

The CPU scheduling policy is Round Robin (RR) with Time Quantum as 2 Units. Calculate the average waiting time and average turnaround time. Also, show the queue maintained for process arrival.

- (c) Explain the convoy effect with an example. [2]